

Enhancing Digital Government and Economy

Course: Software Quality Assurance

Center: DUET

Assignment No: 01

Assignment Name: Comparative Analysis of 10 types Website

Submitted To

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Title: Comparative Analysis of Website Architectures & SQA Testing Strategies

Objective:

The objective of this assignment is to enable students to:

1. Identify and classify 10 distinct types of websites based on their purpose and functionality.
2. Define each website type and list its core functional features.
3. Recognize exclusive SQA challenges (unique bugs, performance risks, security vulnerabilities) associated with each type.
4. Provide real-world examples for each category.
5. Understand how SQA strategies must adapt to different website architectures rather than applying a one-size-fits-all testing approach.

Methodology:

1. E-commerce Website

Definition:

An e-commerce website is a digital platform that enables businesses to sell products or services online. It facilitates product browsing, selection, payment processing, and order fulfillment without physical store interaction.

Key Functional Features:

- Product catalog with search and filter functionality
- Shopping cart for temporary item storage
- Secure payment gateway integration (credit cards, digital wallets)
- User account and order history
- Inventory management system
- Customer reviews and ratings

Exclusive SQA Challenges:

- Payment gateway failures causing double charges or failed transactions
- Inventory inconsistency (showing out-of-stock items as available)
- Cart abandonment data loss during session timeout
- Tax calculation errors across different regions
- Currency conversion bugs for international sales
- Coupon and discount code logic failures

Real-World Examples:

- Amazon (www.amazon.com)
 - eBay (www.ebay.com)
 - Shopify (www.shopify.com)
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2. Social Media Website

Definition:

A social media website is an interactive platform that allows users to create profiles, share content, communicate with others, and build online communities or networks.

Key Functional Features:

- User profile creation and customization
- News feed or timeline displaying curated content
- Like, comment, share, and react functionality
- Private messaging and group chats
- Content upload (images, videos, text)
- Notification system for user interactions

Exclusive SQA Challenges:

- Real-time feed inconsistency (missing or duplicate posts)
- Notification duplication or delay under high load
- Privacy setting leaks (private content becoming visible)
- Cross-site scripting (XSS) vulnerabilities in comments or messages
- API rate limiting failures leading to spam or denial-of-service
- Media upload corruption or orientation errors

Real-World Examples:

- Facebook (www.facebook.com)
 - Instagram (www.instagram.com)
 - X / Twitter (www.twitter.com)
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3. CMS / Blog Website

Definition:

A Content Management System (CMS) or blog website allows users to create, manage, modify, and publish digital content without specialized technical knowledge.

Key Functional Features:

- WYSIWYG (What You See Is What You Get) editor
- Tagging, categorization, and archiving
- Comment section with moderation tools
- Multi-author roles and permissions
- Scheduled publishing and draft management
- SEO tools and permalink customization

Exclusive SQA Challenges:

- Markdown or HTML injection in post content
- Comment spam and bot protection failures
- Broken permalinks after title or URL changes
- Media upload corruption or compression artifacts
- Draft vs. published version mismatch
- Theme or plugin compatibility breaking layout

Real-World Examples:

- WordPress (www.wordpress.com)
- Medium (www.medium.com)
- Blogger (www.blogger.com)

4. Educational / LMS Website

Definition:

An Educational or Learning Management System (LMS) website is a platform designed to deliver, track, and manage online education, training courses, and assessment programs.

Key Functional Features:

- Course catalog and enrollment system
- Video lecture hosting and playback
- Quizzes, assignments, and automated grading
- Progress tracking and completion certificates
- Discussion forums and instructor messaging
- Gradebook and performance analytics

Exclusive SQA Challenges:

- Quiz scoring calculation errors (partial credit, negative marking)
- Video playback compatibility across browsers and devices
- Progress tracking loss after session timeout or browser refresh
- Certificate generation forgery or formatting errors
- Timezone issues for assignment deadlines and live classes
- Accessibility compliance failures (screen reader, keyboard navigation)

Real-World Examples:

- Coursera (www.coursera.org)
- Khan Academy (www.khanacademy.org)
- Moodle (www.moodle.org)

5. Banking / FinTech Website

Definition:

A banking or FinTech website is a secure digital platform that provides financial services including account management, fund transfers, bill payments, investments, and loans.

Key Functional Features:

- Secure user login with multi-factor authentication (MFA)
- Account balance and transaction history
- Fund transfer (internal, external, wire, UPI)
- Bill payment and scheduled payments
- Check deposit via mobile or document upload
- Financial statements and tax documents

Exclusive SQA Challenges:

- Double transaction posting due to retry logic failure
- Session timeout not terminating properly (security risk)
- Interest calculation precision errors (rounding issues)
- Insufficient balance validation bypass
- Man-in-the-middle vulnerabilities on API calls
- Audit log gaps (missing transaction trails)

Real-World Examples:

- Chase (www.chase.com)
- PayPal (www.paypal.com)
- Bank of America (www.bankofamerica.com)

6. Streaming (Video / Audio) Website

Definition:

A streaming website delivers continuous video or audio content over the internet in real-time, allowing users to watch or listen without downloading complete files.

Key Functional Features:

- On-demand or live streaming playback

- Adaptive bitrate streaming (adjusts quality based on connection)
- Play, pause, seek, volume, and fullscreen controls
- Subtitles, captions, and audio language selection
- Watchlist or playlist creation
- Resume playback across devices

Exclusive SQA Challenges:

- Buffering latency and rebuffering events under poor network
- Audio-video sync drift over time
- Subtitle misalignment or missing captions
- Seek bar inaccuracy (resuming at wrong timestamp)
- DRM (Digital Rights Management) license failures
- Memory leaks during extended playback

Real-World Examples:

- Netflix (www.netflix.com)
- YouTube (www.youtube.com)
- Spotify (www.spotify.com)

7. Government / Portal Website

Definition:

A government or portal website is an official digital service platform operated by a public authority to provide information, forms, permits, and citizen services.

Key Functional Features:

- Public information dissemination (laws, policies, announcements)
- Online form submission (tax filing, license renewal)
- Document download (applications, certificates)
- Appointment scheduling for government services
- Status tracking of applications
- Multilingual support

Exclusive SQA Challenges:

- High-traffic load during tax deadlines or disaster events
- Accessibility compliance for citizens with disabilities (Section 508, WCAG)
- Form validation failures leading to rejection of legitimate applications
- Session state loss during long multi-page applications
- Data privacy between citizen submissions
- Broken internal links after site restructuring

Real-World Examples:

- [USA.gov](http://www.usa.gov) (www.usa.gov)
- [GOV.UK](http://www.gov.uk) (www.gov.uk)
- [India.gov.in](http://www.india.gov.in) (www.india.gov.in)

8. Corporate / Brochure Website

Definition:

A corporate or brochure website is a relatively static website that provides company information, brand messaging, product overviews, and contact details without complex transactions.

Key Functional Features:

- About us and company history
- Products or services overview
- Contact information and location map
- Team or leadership profiles
- News or press release section
- Career or job opening listings

Exclusive SQA Challenges:

- Broken contact form submissions (false success messages)
- Responsive design failures on mobile devices
- Cross-browser layout inconsistencies
- Slow-loading high-resolution images
- Newsletter subscription double opt-in failures
- Embedding issues with third-party maps or chatbots

Real-World Examples:

- Apple (www.apple.com)
- Tesla (www.tesla.com)
- Microsoft (www.microsoft.com)

9. Search Engine Website

Definition:

A search engine website allows users to enter queries and retrieves relevant web pages, images, videos, or other content from an indexed database using algorithms.

Key Functional Features:

- Query input with auto-suggestions
- Crawled and indexed web content
- Ranking algorithm for result ordering
- Filters (time, region, content type)
- Spell correction and related queries
- Pagination or infinite scroll of results

Exclusive SQA Challenges:

- Index freshness delay (new content not appearing)
- Ranking accuracy degradation after algorithm update
- Infinite scroll or pagination drift (duplicate or missing results)
- Auto-suggest showing inappropriate or outdated terms
- Image search returning broken or mismatched thumbnails
- Search latency under high query volume

Real-World Examples:

- Google (www.google.com)
- Bing (www.bing.com)
- DuckDuckGo (www.duckduckgo.com)

10. Web Application (SaaS)

Definition:

A Software as a Service (SaaS) web application is a cloud-based software delivered via browser, where users access functionality on a subscription or pay-per-use model without local installation.

Key Functional Features:

- User authentication and role-based access control
- Core business logic (project management, CRM, design, analytics)
- Data storage and synchronization across devices
- Subscription and billing management
- API access for third-party integrations
- Real-time collaboration (multi-user editing)

Exclusive SQA Challenges:

- Multi-tenant data isolation failure (one user seeing another's data)
- API versioning breaking existing integrations
- Subscription plan feature gating errors
- Real-time sync conflicts in collaborative editing
- Data export/import corruption during migration

- Billing cycle miscalculations for prorated subscriptions

Real-World Examples:

- Salesforce (www.salesforce.com)
- Trello (www.trello.com)
- Canva (www.canva.com)

Conclusion:

This assignment provided a structured analysis of ten distinct website types from a Software Quality Assurance (SQA) perspective. By moving beyond simple visual or functional classification, the exercise emphasized that each website type carries unique quality risks, testing priorities, and failure modes that a generic testing strategy would fail to address.